

MATH 2228-601, SUMMER II 2025, MOCK MIDTERM 1

INSTRUCTIONS

1. Do not open the exam paper or start working until you are told to do so.
2. Bubble in your name as DOE John S (if it is John S. Doe), bubble in your student ID number.
3. NO NOTES OR BOOKS ARE ALLOWED. You may not communicate with others during the exam. Please be considerate and refrain from making noise. Switch off any beepers, cell phones, etc.
4. Bubble in your answer to, say, Question 4, on line 4.
5. The exam is self-explanatory. The instructor will not interpret the questions for you. You may, however, ask for another copy of the exam if your copy is illegible.
6. Have your calculator in **RADIAN MODE**. You can set radian mode by pressing a button called MODE, SET UP, or DRG (degree, radian, gradian).

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1. Let Q_1 be the frequency of cat, Q_2 the frequency of dog, Q_3 the frequency of mouse, and Q_4 the frequency of rat in the list below.

rat, rat, dog, cat, dog, mouse, cat, mouse, dog, rat, mouse, rat, mouse, mouse, dog, dog, mouse, rat, mouse, rat, rat, mouse, cat, cat, mouse, rat, mouse, rat, mouse, rat, cat, cat, cat, dog, mouse, cat, rat, cat, mouse, cat

Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3| + 4|Q_4|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

2. Let Q_1 be the relative frequency of cat, Q_2 the relative frequency of dog, Q_3 the relative frequency of mouse, and Q_4 the relative frequency of rat in the list below.

rat, mouse, cat, rat, dog, dog, mouse, rat, rat, rat, mouse, mouse, dog, cat, cat, cat, rat, mouse, cat, cat, rat, mouse, mouse, rat, rat, rat, cat, dog, cat, dog, rat, mouse, dog, dog, rat, cat, rat, rat, mouse, mouse

Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3| + 4|Q_4|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

3. Let $x_i = 40 \sin^2(351i)$ for $i = 1, \dots, 10$. Let Q_1 be the frequency of x_i falling into the interval $(0, 13]$, Q_2 be the frequency of x_i falling into the interval $(13, 18]$, Q_3 be the frequency of x_i falling into the interval $(18, 26]$, and Q_4 be the frequency of x_i falling into the interval $(26, 31]$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3| + 4|Q_4|)$. Then $T = 5 \sin^2(100Q)$

satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

4. Let $x_i = 40 \sin^2(105i)$ for $i = 1, \dots, 10$. Let Q_1 be the relative frequency of x_i falling into the interval $(0, 10]$, Q_2 be the relative frequency of x_i falling into the interval $(10, 18]$, Q_3 be the relative frequency of x_i falling into the interval $(18, 27]$, and Q_4 be the relative frequency of x_i falling into the interval $(27, 35]$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3| + 4|Q_4|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

5. Let Q_1 be the median of the numbers $x_i = 40 \sin^2(420i)$ for $i = 1, \dots, 10$. Let $Q = \ln(3 + |Q_1|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

6. Let Q_1 be the mean of the numbers in the list below.

151, 173, 100, 60, -22, 95, 103, 0, 168, 17, 101, -7, 10, 184, -12, 83, 163, 112, 191, 12

Let $Q = \ln(3 + |Q_1|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

7. Let Q_1 be the mean of the numbers given by f_i copies of x_i listed below as pairs (x_i, f_i) .

(65, 54), (-82, 66), (-15, 85), (98, 90), (-22, 69), (-39, 60), (43, 64), (-38, 44), (-59, 28), (-16, 17)

Let $Q = \ln(3 + |Q_1|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

8. Let Q_1 be the sample mean and Q_2 the sample standard deviation of the numbers given by f_i copies of x_i listed below as pairs (x_i, f_i) .

(81, 70), (-27, 36), (46, 76), (37, 69), (4, 76), (-66, 79), (-56, 36), (-95, 48), (-36, 38), (-86, 21)

Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

9. Let Q_1 be the sample mean and Q_2 the sample standard deviation of the numbers listed below.

12.88, -8.52, -6.83, 8.72, 13.63, 0.05, -5.55, 17.34, 18.83, 0.63, 18.21, 3, -2.01, 12.9, 1.69, 1.02, -1.44, 19.19, -2.78, -0.09

Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

10. Let Q_1 be the weighted GPA (grade point average) rounded to two decimals after the point of a college student for the semester given the classes below. Note that A corresponds to 4 points, B to 3, C to 2, D to 1, F to 0.

Class 1, grade B, credit 4; Class 2, grade A, credit 3; Class 3, grade F, credit 3; Class 4,

grade C, credit 2; Class 5, grade D, credit 5

Let $Q = \ln(3 + |Q_1|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

11. Let Q_1 be the sample mean rounded to two decimals after the point and Q_2 the sample standard deviation rounded to two decimals after the point of the salary at a company where 278 people earn in the bracket $(30000, 45000]$, 262 people earn in the bracket $(45000, 58000]$, 249 people earn in the bracket $(58000, 68000]$, 238 people earn in the bracket $(68000, 77000]$, 223 people earn in the bracket $(77000, 96000]$, which accounts for everybody. Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

12. Suppose a distribution has mean $\mu = 341$ and standard deviation $\sigma = 51.1$. Let Q_1 be the value which belongs to the z-score -1.2 , Q_2 the value which belongs to the z-score -1.8 , Q_3 the z-score of the value 187.7, and Q_4 the z-score of the value 203.03. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3| + 4|Q_4|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.